MIL–G–17927A
7 JANUARY 1966
SUPERSEDING
MIL–G–17927(NAVY)
29 APRIL 1955
(See 6.3)

MILITARY SPECIFICATION

GASKETS, ASBESTOS–METALLIC COVER, RUBBER CORE

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers gaskets for use on hinged closures where flame resistance is required.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

HH–P–31 — Packing Material, Asbestos, Metallic Cloth, Sheet and Tape.

PPP–B–566 — Boxes, Folding, Paperboard.

PPP–B–636 — Box, Fiberboard.

PPP–B–640 — Boxes, Fiberboard, Corrugated, Triple-Wall.

PPP–B–676 — Boxes, Set-Up, Paperboard.

MILITARY

MIL–P–116 — Preservation, Method of.

STANDARDS

FEDERAL

FED–STD–601 — Rubber: Sampling and Testing.

MILITARY

MIL–STD–129 — Marking for Shipment and Storage.

MIL–STD–407 — Visual Inspection Guide for Rubber Molded Items.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification

FSC 5330

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to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids, shall apply.

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules.

(Application for copies should be addressed to Official Classification Committee, 1 Park Avenue at 33rd Street, New York, N. Y. 10016.)

3. REQUIREMENTS

3.1 Material and construction. The gaskets shall consist of a rubber core covered with two plies of asbestos-metallic cloth or two successive asbestos-metallic braids. The plies or braids shall be adhered together and to the rubber core with a nonflammable adhesive.

3.1.1 *Asbestos-metallic cloth.* The asbestos-metallic cloth shall conform to HH–P–31, except that it shall be flame-resistant and the cloth shall weigh not less than 3½ pounds per square yard.

3.1.2 *Asbestos-metallic braids.* The asbestos metallic braids shall be constructed from two strands, each consisting of two asbestos yarns and two brass wires. The brass wire shall be 0.007 inch in diameter plus or minus 0.001 inch. The yarns shall contain not less than 90 percent of asbestos fiber and shall weigh not less than 1 pound per 1,000 yards. The braid shall be so constructed that there will be a minimum of 16 strands per inch in either direction, measured at right angles to the direction of the strands. The inner braid shall be bonded to the rubber core. The inner and outer braid shall be bonded with a nonflammable adhesive. The outer braid shall be treated to keep down lint so that it will not detract from the flame resistance of the gasket.

3.1.3 *Rubber core.* The rubber core material shall conform to the physical requirements specified in table I.

TABLE I. *Physical requirement of rubber.*

	Requirement	Test procedure
Initial properties:		
Tensile strength, psi, minimum	1000	4.4.2
Ultimate elongation, percent, minimum	300	4.4.2
Hardness, durometer points	45 ± 5	4.4.3
Specific gravity, maximum	1.40	4.4.4
Properties after oven aging for 46 ± ½ hours at 194° ± 2°F:		
Tensile strength, percent of initial, minimum	75	4.4.5.1
Ultimate elongation, percent of initial, minimum	70	4.4.5.1
Hardness, Shore A durometer, maximum	55	4.4.5.2
Compression set, percent, maximum	40	4.4.5.3
Delamination after liquid immersion	no delamination	4.4.6

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3.1.4 *Adhesion.* The nonflammable adhesive used between the inner and outer cover and between the inner cover and the rubber core shall give such adhesion that it will not be possible to strip the inner and outer asbestos-metallic cloth or braid cleanly from each other or from the rubber core when tested as specified in 4.4.7.

3.2 Flame and heat resistance. The finished gasket shall satisfactorily pass the test specified in 4.4.8.

3.3 Dimensions and tolerances. Unless otherwise specified (see 6.1), the dimensions for the rubber core and the finished gasket shall be as shown in table II. The finished gasket shall be within a tolerance of plus 1/16 inch. No minus tolerance will be permitted.

TABLE II. *Dimensions.*

Rubber core		Finished gasket		
Thickness (min.)	Width (min.)	Thickness (T) (min.)	Width (W) (min.)	Length
Inch	Inch	Inch	Inches	Feet
½	½	¾	¾	11–23
½	1	¾	1-¼	11–23

3.4 Workmanship. The workmanship of the finished gaskets shall be such as to meet all applicable requirements of this specification. The finished gaskets shall be examined as specified in 4.3.1.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 *Lot.* For the purposes of sampling, examination, and tests, a lot shall consist of all gaskets of the same size, produced in one plant under essentially the same conditions, not exceeding 2500 pounds, and offered for delivery at one time.

4.2.2 *Sampling for examination.* A random sample of gaskets shall be taken from each lot in accordance with the sampling plan given in table III for the examination specified in 4.3.1.

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TABLE III. *Sampling for examination.*

Lot size Number of gaskets	Sample size Number of gaskets	Number of nonconforming or defective gaskets	
		Acceptance number	Rejection number
Up to 5	all	0	1
6 to 50	5	0	1
51 to 150	20	1	2
151 to 280	32	2	3
281 to 500	50	3	4
501 and over	80	5	6

4.2.3 *Sampling for tests.* Random samples shall be taken from each lot that passes the examination of 4.3.1, in sufficient quantity to conduct either the production check tests or the quality conformance tests specified in 4.3.2 or 4.3.3, as applicable. Since the items are of such size and shape that all test specimens cannot be prepared from them, substitute samples shall be furnished with each lot in the form of two sample pieces 6 inches by 6 inches by 0.080 plus or minus 0.010 inch thick and one sample piece approximately 4 inches by 3 inches by 0.500 plus or minus 0.010 inch thick. The substitute samples shall be certified to be of the same material and equivalent cure as that used in the lot of finished material offered for delivery.

4.2.3.1 *Special sample for tests.* A square-yard sample of the asbestos-metallic cloth or three linear yards of the asbestos-metallic braid, as used, shall be furnished by the manufacturer with each lot for tests specified in 4.4.9. The samples of cloth or braid shall be identical to that used in the lot of finished material.

4.3 Quality conformance inspection.

4.3.1 *Examination.* Each of the samples taken in accordance with 4.2.2 shall be examined visually for workmanship, appearance, dimensions, and tolerances. MIL–STD–407 shall be used to determine and evaluate visual defects.

4.3.1.1 *Rejection.* Any gasket in the sample containing one or more visual or dimensional defects shall be subject to rejection and if the number of defective or nonconforming gaskets exceeds the applicable acceptance number of table III, this shall be cause for rejection of the entire lot represented by the sample.

4.3.2 *Production check tests.* Production check tests shall be conducted on samples from (or representing) the first lot of material and from every tenth lot thereafter. All the tests specified in 4.4 shall be conducted.

4.3.3 *Quality conformance tests.* Quality conformance tests shall be conducted on samples from (or representing) all intermediate lots on which production check tests were not performed. The tests specified in 4.4.2, 4.4.3, 4.4.5.3, 4.4.8, and 4.4.9 shall be conducted.

4.3.4 *Action in case of nonconformance.* If any of the samples in 4.3.2 and 4.3.3 is found not to be in conformance with the requirements of this specification, this shall be cause for rejection of the lot represented by the sample. Furthermore, if production check test fails, additional samples shall be taken from each subsequent lot and shall be subjected to the test or tests wherein the failure occurred. Each lot shall be considered acceptable only after satisfactory results are obtained on the test or tests by all the samples taken to represent the lot. This additional testing shall be discontinued after four successive lots have passed the test or tests.

4.4 Test procedures.

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4.4.1 *Testing conditions.* Unless otherwise specified (see 6.1), the testing conditions shall be in accordance with the temperature and humidity of conditioning room requirements of FED–STD–601.

4.4.2 *Tensile strength and ultimate elongation.* The tensile strength and ultimate elongation shall be determined by methods 4111 and 4121, respectively, of FED–STD–601. Die III specimens shall be used for all tensile determinations.

4.4.3 *Hardness.* Hardness shall be determined with a Shore A durometer on 0.500 plus or minus 0.010 inch thick specimens in accordance with method 3021 of FED–STD–601. The 3-second reading shall be taken.

4.4.4 *Specific gravity.* The specific gravity shall be determined by method 14011 of FED–STD–601.

4.4.5 *Oven aging.* Method 7221 of FED–STD–601 shall be used for oven aging the test specimens as specified for the individual tests for 46 plus or minus ½ hours at 90° ± 1.1°C. (194° ± 2 F.)

4.4.5.1 *Tensile properties after oven aging.* The tensile properties after oven aging shall be determined by the procedure in 4.4.2.

4.4.5.2 *Hardness after oven aging.* The hardness after oven aging shall be determined by the procedure in 4.4.3.

4.4.5.3 *Compression set after oven aging.* Compression set shall be determined in accordance with method 3311 of FED–STD–601, with specimens clamped to 40 percent deflection and aged in accordance with 4.4.5.

4.4.6 *Delamination.* The delamination test shall be conducted in accordance with method 6311 of FED–STD–601.

4.4.7 *Adhesion test.* The inner and outer asbestos-metallic cloth or braid shall be stripped from the rubber core and from each other without the use of solvents. The asbestos-metallic cloth or braid shall be examined after stripping apart.

4.4.8 *Flame and heat resistance.* A specimen of the finished gasket shall be held for 15 seconds with one face across the flame of a Tirrill type burner. The lower side of the gasket shall be held even with the tip of the blue cone of the flame. The temperature of the flame, measured with a pyrometer, shall be approximately 870°C. (1,600°F) at the tip of the blue cone. If a pyrometer is not available for measuring the temperature, the tip of the blue cone of the flame shall be not less than 1 inch or more than 1½ inches high. The specimen, when cut open at the exposed section, shall not show any indication of melting of the rubber core, nor shall the adhesive compound have been consumed. In addition, a specimen of the gasket shall be held, for not more than 60 seconds, over a 870°C. (1,600°F) flame, as specified herein, until the rubber core possibly ignites or bursts into flame. The gasket, if burning, shall not continue to blaze when removed from the burner flame for more than 60 seconds. Cloth or braid covering shall hold its shape and texture.

4.4.9 *Tests on asbestos-metallic cloth or asbestos-metallic braid.* The sample of asbestos-metallic cloth or asbestos-metallic braid shall be tested in accordance with HH–P–31 to determine conformance of the asbestos-metallic cover with the requirements of this specification (see 3.1.1 and 3.1.2).

4.5 Examination of preparation for delivery. A sample pack from each lot shall be examined for compliance with section 5 of this document.

5. PREPARATION FOR DELIVERY

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5.1 Packaging. Packaging shall be level A or C as specified (see 6.1).

5.1.1 *Level A.* Unless otherwise specified (see 6.1), the gaskets shall be individually packaged in folding or set-up boxes conforming to PPP–B–566 and PPP–B–676, respectively. Packaging shall meet the requirements of method III of MIL–P–116.

5.1.2 *Level C.* Packaging of gaskets shall be sufficient to afford adequate protection against deterioration and physical damage during shipment from the supply source to the using activity and until early use. Packaging may conform to the supplier's commercial practice when such meets the requirements specified.

5.2 Packing. Packing shall be Level A, B, or C as specified (see 6.1).

5.2.1 *Level A.* Gaskets, packaged as specified (see 6.1), shall be packed for shipment in containers conforming to class weather resistant of PPP–B–636, or class 2 of PPP–B–640, at the option of the supplier. All corners and edge seams and manufacturer's joints of fiberboard boxes shall be waterproofed with tape in accordance with the appendix to the box specification. Boxes shall be closed and banded with pressure sensitive tape in accordance with the applicable box specification or appendix thereto. The gross weight of fiberboard boxes shall not exceed the weight limitations of the applicable fiberboard box specification.

5.2.2 *Level B.* Gaskets, packaged as specified (see 6.1), shall be packed for shipment in containers conforming to class domestic of PPP–B–636, or class 1 of PPP–B–640, at the option of the supplier. Box closures shall be as specified in the applicable box specification or appendix thereto. The gross weight of the containers shall not exceed the weight limitations of the applicable box specification.

5.2.3 *Level C.* The gaskets shall be packed in a manner which will insure acceptance by common carrier and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early use. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the supplier's commercial practice when such meets the requirements specified.

5.3 Marking. In addition to any special marking required (see 6.1), interior packages and exterior shipping containers shall be marked in accordance with MIL–STD–129 and shall also include the cure date (month and year).

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Dimensions required (see 3.3).

(c) Level of packaging and level of packing required (see 5.1 and 5.2).

(d) Special marking required (see 5.3).

6.2 Precaution. Gaskets should not be lubricated with graphite or other materials.

6.3 Changes from previous issue. The extent of changes (deletions, additions, etc.) preclude the annotation of the individual changes from the previous issue of this document.

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Custodians:

Army—MR

Navy—SH

Air Force—69

Review activities:

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Air Force—69, 85

User activities:

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Preparing activity:

Navy—SH

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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